

Omni2LOD3: Extracting Façade Geometry and Semantic Features from Omnidirectional Imagery and Drone Photogrammetry

Keywords: LOD3, Building reconstruction, CityGML, Drone photogrammetry, Omnidirectional imagery, 3D point cloud

Abstract

Generating Level of Detail 3 (LOD3) building models requires accurate façade representation. However, existing workflows often rely on LiDAR, dense multiview imagery, or pre-existing lower-LOD models, limiting their applicability in data-scarce environments. Most methods are also developed on buildings with simple façade geometry, focusing mainly on detecting openings while keeping features simplified. This study presents Omni2LOD3, a semi-automated workflow that extracts LOD3-ready geometric and semantic components from UAV photogrammetry and a limited number of omnidirectional images, challenging the assumption that façade-level reconstruction requires dense multiview stereo or LiDAR acquisition. The workflow is developed on a building containing curved façade regions, overhangs, and partial occlusions that are typically difficult to reconstruct using sparse image-based methods. An LOD2 base model is derived from nadir drone imagery. Façade geometry and semantic features are extracted from omnidirectional images using monocular depth estimation and color-based segmentation, establishing a lightweight baseline for semantic extraction from sparse image-derived point clouds. Geometric evaluation against a SLAM-based terrestrial LiDAR benchmark yielded a mean cloud-to-cloud distance of 0.46 m, sufficient to capture façade structure at LOD3, where planar consistency is prioritized over fine-scale precision. Semantic evaluation shows that façade element recoverability is class-dependent, with windows and building installations detected more reliably, while doors remained challenging due to limited visual separability. After manual refinement, detection completeness reached 81%. Overall, results demonstrate the feasibility of recovering meaningful façade geometry and semantics from sparse omnidirectional imagery, with these outputs serving as the foundation for CityGML-compliant LOD3 generation within the broader Omni2LOD3 pipeline.

1. Introduction

3D city models are an important component of modern geospatial infrastructure and are widely used in applications such as urban planning, environmental simulation, and digital twins. The usefulness of these models depends on their Level of Detail (LOD), which determines their geometric and structural complexity (Biljecki et al., 2016). In CityGML 3.0, Levels of Detail describe geometric representation rather than semantic completeness, meaning that façade element recoverability contributes to LOD3 readiness even when semantic reconstruction remains partial. Recent updates further emphasize that LOD is defined in terms of spatial representation rather than semantic decomposition, where elements such as windows and doors are treated separately (Kutzner et al., 2020). Within this framework, LOD3 building models represent the highest level of exterior geometric detail typically used in façade-level reconstruction workflows (Open Geospatial Consortium (OGC), 2021), as illustrated in Figure 1.

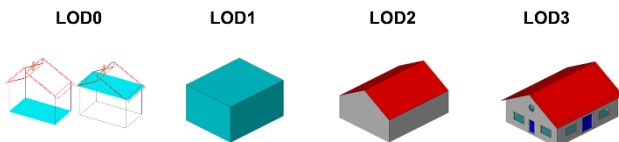


Figure 1. Representation of the same real-world building in different Levels of Detail. Adapted from OGC (2021)

Traditionally, LOD3 building reconstruction relies on LiDAR data, which provides accurate 3D measurements of building geometry but requires costly acquisition and specialized workflows (Wang, 2013). In addition, LiDAR-based pipelines

are not always feasible in many regions, particularly where national-scale LiDAR surveys or detailed 3D building datasets are unavailable.

Image-based reconstruction methods have therefore gained traction as an alternative for generating 3D building models. Unmanned Aerial Vehicle (UAV) photogrammetry has emerged as a widely used technique for capturing building geometry due to its flexibility and relatively low cost. UAV imagery can provide high-resolution observations of roofs and building structures, enabling detailed reconstruction (Vacca et al., 2017). However, aerial imagery alone often struggles to capture façade information due to limited viewing angles, occlusions, or distortions, particularly in dense urban environments with surrounding structures or large canopies. As a result, ground-level details and recessed features are often incomplete. To address the limitations of aerial photogrammetry, close-range dense image capture offers an alternative, as photogrammetric geometry reconstruction techniques from multiple overlapping images reliably produce accurate point clouds (Pantoja-Rosero et al., 2022). However, the need for several overlapping image captures normally requires extensive field time.

To address these limitations, recent studies have integrated complementary image sources that provide additional perspectives on building façades. Panoramic or omnidirectional imagery provides wide-field-of-view observations of urban scenes and has shown potential for improving façade reconstruction (Ogawa et al., 2024; Tang et al., 2025). These perspectives complement aerial observations by providing direct views of vertical surfaces while reducing the number of required images. Their wide field of view also enables more complete

façade coverage, even when captured from fewer or varying distances.

This study introduces Omni2LOD3, a semi-automated workflow for generating LOD3-ready geometric and semantic façade components from sparse omnidirectional imagery and UAV-derived base models. The methodological contribution of this study is to demonstrate that façade-level geometry suitable for semantic structuring can be reconstructed from a sparse set of omnidirectional images using monocular depth estimation. These reconstructions can be integrated with UAV-derived LOD2 data to produce CityGML-compatible LOD3-ready components without dense multiview stereo or LiDAR acquisition. Rather than presenting a complete end-to-end LOD3 pipeline, the proposed approach focuses on recovering structurally meaningful façade geometry and semantic elements under constrained acquisition conditions, establishing a lightweight baseline for semantic extraction from image-derived façade point clouds.

2. Related Work

Research on 3D building reconstruction has produced diverse workflows across different Levels of Detail. To situate the present study, the following sections review studies relevant to LOD3 reconstruction, image-based modeling, omnidirectional imagery, and façade element extraction.

2.1 LOD2 and LOD3 Reconstruction Pipelines

Existing workflows follow two main strategies, namely reconstruction from raw data and refinement of lower-LOD models. From-scratch methods typically rely on dense image matching or point clouds. For example, Pantoja-Rosero et al. (2022) generate an LOD2 model from multiview imagery and subsequently derive LOD3 by detecting façade openings and projecting them onto the reconstructed model. Other approaches refine pre-existing lower-LOD models. Wysocki et al. (2024) enhance LOD1/LOD2 models using MLS point clouds, while Texture2LoD3 enriches models using panoramic imagery and façade segmentation rather than reconstructing geometry (Tang et al., 2025). Omni2LOD3 applies a fusion of both strategies, featuring a from-scratch LOD2 reconstruction followed by façade-level refinement towards LOD3.

Multi-sensor workflows are also common. Gruen et al. (2019) integrate UAV imagery, aerial LiDAR, and mobile LiDAR, along with procedural modeling and interactive editing, to generate semantically enriched LOD3 models. Harshit et al. (2024), on the other hand, combine LiDAR, UAV photogrammetry, BIM, and IFC-to-CityGML conversion. Despite these advances, many pipelines depend on LiDAR, dense imagery, BIM conversion, or existing LOD2 models. In contrast, our study investigates whether both façade geometry and semantics can be derived under more constrained data conditions using less expensive and more accessible data.

2.2 Image vs LiDAR-based reconstruction

Many LOD3 pipelines rely on LiDAR due to its ability to directly capture accurate 3D geometry. However, beyond acquisition cost, workflows require specialized hardware and processing pipelines, which can limit scalability and accessibility (Wang, 2013). UAV photogrammetry offers a lower-cost alternative. Nadir imagery captures roofs effectively, while oblique imagery improves façade completeness (Vacca et al., 2017). However, this method faces challenges when capturing vertical surfaces,

particularly in occluded environments. Ogawa et al. (2024) note that aerial observations are effective for roofs and footprints but are less suitable for detailed façade reconstruction due to perspective and resolution constraints. Given these limitations, combining aerial and ground-level observations provides a natural approach to improving 3D building model generation, as the two perspectives offer complementary views of its geometry. Our work follows this direction but distinguishes itself by operating under sparse acquisition constraints, demonstrating that meaningful façade geometry can be recovered without dense ground-level capture.

2.3 Omnidirectional Imagery for Façade Geometry

Omnidirectional imagery complements aerial data by capturing wide façade views from close range and has been used for façade analysis and LOD3 enrichment. Ogawa et al. (2024) combine omnidirectional street images with building footprints and aerial photographs to generate LOD3 models. Similarly, Texture2LoD3 (Tang et al., 2025) uses panoramic street-view imagery to enrich existing models through segmentation rather than reconstructing façade geometry. Panoramic imagery can also reduce the number of images required for façade capture due to its 360° field of view (Gao et al., 2010). This is especially relevant for workflows that may be developed to use publicly available street-level imagery, where image density is limited.

Recent work also explores its use for geometric reconstruction. Saura-Herreros et al. (2021) estimate depth from panoramic images using cubemap projection and monocular depth estimation, while Deng et al. (2025) demonstrate real-time omnidirectional depth mapping using multiview fisheye systems. In contrast to these approaches, Omni2LOD3 reconstructs façade geometry directly from sparse omnidirectional images rather than using them solely for texture mapping or semantic enrichment.

2.4 Façade Segmentation Methods

Façade feature extraction methods generally fall into two categories, namely image-based semantic segmentation and point-cloud-based classification. Image-based methods typically use convolutional neural networks (CNNs) to detect components such as windows and doors (Tang et al., 2025; Wysocki et al., 2024). CNNs are specialized deep learning models that train on annotated data. However, they work for simple façades with few occlusions and plain architecture. For buildings with more complex geometry, more robust models are needed. On the other hand, point-cloud-based approaches use geometric features and spatial relationships to identify façade elements. These methods are commonly developed for LiDAR-derived datasets, where high point density supports reliable feature extraction (Gruen et al., 2019; Wysocki et al., 2024). These methods often rely on surface normals and planar constraints, which may be less effective for irregular or complex architectural features.

Image-derived point clouds retain radiometric attributes such as RGB color, which remain underutilized in many segmentation workflows (Zhan et al., 2009). Prior work has explored color-based segmentation for urban-scale analysis (Zhai et al., 2023), and color is also used as an input feature in learning-based models (Zhang et al., 2022). This demonstrates the potential of color-based segmentation for façade analysis, particularly where geometric cues are insufficient. In this context, color-based clustering is explored as a simple and interpretable approach for extracting façade elements from sparse image-derived point clouds, positioning it as a starting point for semantic extraction.

3. Methodology

The Omni2LOD3 workflow integrates nadir drone photogrammetry and a limited number of ground-level omnidirectional images to generate LOD3-ready geometric and semantic components. As illustrated in Figure 2, the pipeline first reconstructs an LOD2 base model from UAV imagery, then extracts and semantically segments façade point clouds from omnidirectional images. Each stage is designed to operate with limited data, prioritizing accessible acquisition and user-guided processing over approaches that require dense inputs or extensive annotation. Together, these outputs establish the geometric and semantic foundation for CityGML LOD3 reconstruction within the broader Omni2LOD3 framework.

3.1. Study Area

The study was conducted within a university campus in the Philippines. A multi-story academic building was selected as the primary development site. The building contains both simple planar façades and complex architectural features such as curved wall segments, recessed elements, and partial occlusions from surrounding vegetation. These characteristics provide a more realistic test case for façade extraction than many existing LOD3 studies, which often evaluate buildings with relatively simpler geometries, characterized by planar surfaces and minimal occlusions (Harshit et al., 2024; Pantoja-Rosero et al., 2022; Tang et al., 2025; Wysocki et al., 2024).

3.2. Data

UAV and omnidirectional imagery were the two primary datasets used in this study. Nadir images captured with a DJI Mini 3 Pro drone were used to reconstruct the initial building geometry. These images provide high-resolution observations of roof structures required for LOD2 generation. Ground-level omnidirectional images were captured using an Insta360 X4 camera mounted approximately 1.3 m above ground. Each image provides a full 360° field of view, allowing large portions of façades to be observed from relatively few locations. A total of 13 images were collected around the building, with positions selected based on visibility rather than fixed spacing. Given the

building’s approximate 176-meter perimeter, this represents a sparse acquisition setup compared to typical multiview stereo requirements. A SLAM-based terrestrial LiDAR dataset of the study building, collected by Ingles et al. (2024) using a Foxtech Simultaneous Localization and Mapping (SLAM) 100 scanner, was used for accuracy assessment. It was selected as the geometric reference due to its dense, high-accuracy 3D measurements, making it an appropriate benchmark for evaluating image-derived point clouds (Wang, 2013). Architectural plans were also used as supplementary references for façade dimensions, element counts, and structural features.

3.3 LOD2 Model Generation

3.3.1 Nadir Point cloud generation. Nadir UAV images were processed in Pix4Dmapper using Structure-from-Motion (SfM). This method reconstructs scene geometry by identifying tie points across overlapping images, estimating camera poses, and generating a dense point cloud through bundle adjustment (Westoby et al., 2012). SfM was selected for its well-documented reliability with UAV imagery, where high image overlap and consistent viewing angles support accurate tie-point matching and dense reconstruction. The resulting point cloud provides detailed observations of roof structures and upper building surfaces.

3.3.2 Point cloud preprocessing. Since the dataset consists only of nadir imagery, automated building-point classification is unreliable. Existing classification methods rely on vertical surface detection and spatial relationships, which are difficult to infer from nadir-only data (Yastikli and Cetin, 2021; Zhang and Zhang, 2017). Manual digitization of the building footprint in QGIS was therefore adopted, using the roof point cloud and the UAV orthomosaic as a reference. This approach was preferred over automated methods, given the nadir-only acquisition setup, where automated segmentation would introduce greater uncertainty than controlled user-guided extraction. The digitized polygon was then used as a spatial mask to isolate building points using a custom Python script. This served a dual purpose, enabling building-point classification and providing the base geometry for LOD2 reconstruction.

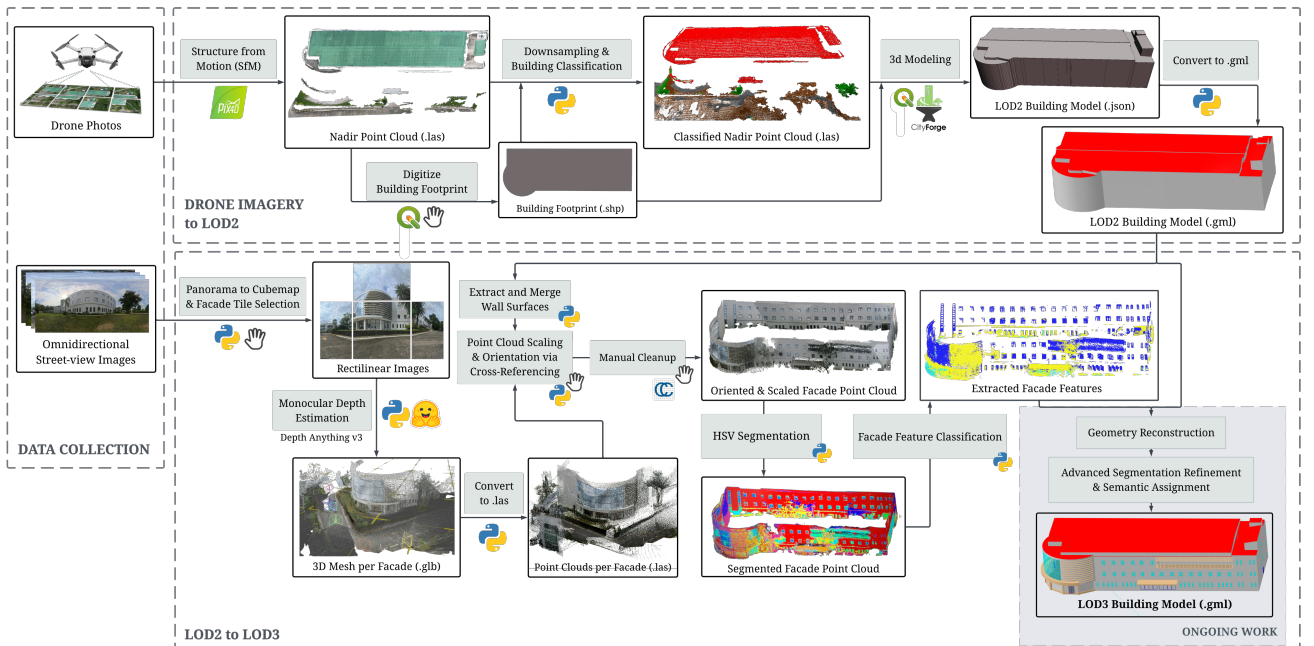


Figure 2. Overview of the Omni2LOD3 pipeline.

3.3.3 LOD2 model reconstruction. The classified point cloud and footprint were processed using the CityForge plugin for QGIS, which reconstructs building envelopes by detecting roof planes and generating wall surfaces (Oslandia, n.d.). CityForge was selected for its streamlined, user-accessible workflow for LOD2 reconstruction within a GIS environment, consistent with the pipeline’s low-cost design. The resulting model is generated in CityJSON format by default. To enable compatibility with downstream processing and CityGML-based reconstruction, the model was converted to CityGML using the citygml4j conversion toolkit (Nagel and Deininger, 2026).

3.4 Façade Geometry and Semantic Feature Extraction

The second stage of the workflow extracts façade-level geometric and semantic information from omnidirectional imagery, as a preliminary step toward LOD3 reconstruction. The output of this stage is a semantically segmented point cloud containing windows, doors, and building installations.

3.4.1 Façade extraction. Omnidirectional images were converted into planar images using the panorama-to-cubemap tool by Crane (2023), which applies Lanczos interpolation (Zhang et al., 2020) to preserve image quality during reprojection. A cubemap projection was adopted because monocular depth estimation models assume a standard perspective projection. Converting to rectilinear tiles eliminates the radial distortions inherent in equirectangular formats that would otherwise degrade depth estimation accuracy. The program allows perspective tilting, ensuring each façade is captured from ground to roof in a single tile, improving depth stitching. Given the sparse dataset, manual tile selection was preferred to reduce uncertainty in façade boundary definition.

3.4.2 Façade point cloud generation. Façade geometry was reconstructed using Depth Anything v3: Recovering the Space from Any Views (Lin et al., 2025). This monocular depth estimation model was selected for its ability to infer scene geometry from images without requiring known camera poses or image overlap, making it well-suited to the sparse, uncalibrated acquisition setup. Each façade was processed individually to maintain geometric consistency within each tile, producing a sparse mesh in an arbitrary coordinate system. The mesh (GLB) is then converted to LAS format, followed by manual cleaning to remove non-façade points. This process produces point clouds representing the geometric structure observed in the omnidirectional imagery.

3.4.3 Point cloud georeferencing. Since the façade point clouds are generated in arbitrary coordinate systems, the LOD2 model was used to align them to a common coordinate system. This approach was adopted because the LOD2 model already occupies a consistent coordinate system derived from the nadir UAV reconstruction, avoiding the need for additional control points or external survey data and maintaining the low-cost design of our workflow. A GUI-based program allows users to select corresponding points between the LOD2 model and each façade point cloud, as visualized in Figure 3. A minimum of three points is required, with snapping to edges and corners to reduce user error.

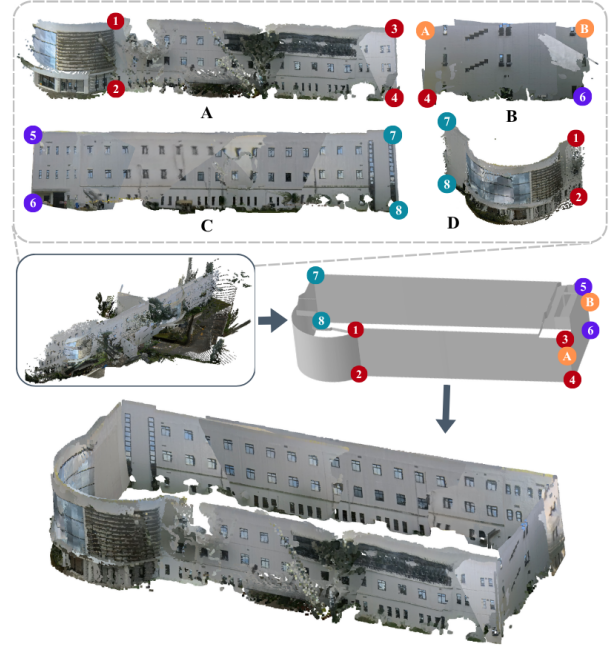


Figure 3. Georeferencing and alignment of façade point clouds

The transformation is computed via the Kubsch-Umeyama algorithm (Lawrence et al., 2019), which estimates scale (s), rotation matrix (R), and translation vector (T) parameters to minimize the root-mean-squared deviation between the ground coordinates (Y) and the arbitrary coordinates (X) as shown in equation 1.

$$Y = sRX + T \quad (1)$$

3.4.4 Semantic feature extraction. A Hue, Saturation, and Value (HSV)-based segmentation approach was applied to differentiate façade elements. Given the sparse and image-derived nature of the façade point cloud, conventional LiDAR-based or deep learning-based semantic segmentation approaches are not directly applicable. HSV-based clustering was therefore adopted as a lightweight and interpretable baseline for semantic extraction under constrained data conditions. Unlike LiDAR-based or learning-based approaches that rely on dense point clouds or annotated training data, this method operates directly on image-derived point clouds with minimal preprocessing. In this context, HSV-based clustering provides a geometry-independent semantic cue that remains stable even when surface normals and planar features are unreliable in sparse, monocular depth-derived façade point clouds. HSV color space was used instead of RGB to reduce sensitivity to illumination and shadow variations across façade surfaces.

Hue is defined as the angular position of a color on the color wheel. Saturation is derived from the normalized difference between the maximum and minimum RGB values, indicating the color intensity. Value represents brightness and is taken as the maximum of the RGB components. The formulas for HSV conversion are given by Equations 2 to 4.

$$H = 60^\circ \times \frac{G - B}{\max(R, G, B) - \min(R, G, B)} \times (\text{mod } 6) \quad (2)^1$$

$$S = \frac{\max(R, G, B) - \min(R, G, B)}{\max(R, G, B)} \quad (3)$$

¹ The hue formula is if Red is the maximum value in R,G,B

$$V = \max(R, G, B) \quad (4)$$

After conversion, K-Means clustering (Sinaga and Yang, 2020) groups points with similar HSV properties. The user then assigns CityGML semantic classes (WallSurface, Door, Window, and Building Installation) onto the resulting clusters. WallSurface points were discarded, as they are already represented in the LOD2 model.

4. Results and Discussion

The intermediate outputs of Omni2LOD3 were evaluated through two complementary assessments. The first involved geometric comparison of the omnidirectional image-derived façade point cloud against a SLAM-based terrestrial LiDAR reference. The second involved semantic evaluation of the segmented façade elements against a manually prepared benchmark.

4.1 Reference Data

The LiDAR point cloud shown in Figure 4 served as the primary geometric reference for evaluating façade point cloud quality.

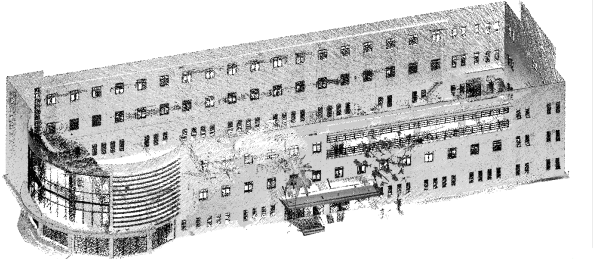


Figure 4. SLAM-based Terrestrial LiDAR point cloud of the study building.

For semantic evaluation, a benchmark model of the building was manually constructed using SketchUp and CityEditor, following a point-cloud-based LOD3 modeling workflow adapted from the Technical University of Munich (2023). Manual construction ensured accurate representation of complex architectural features, including curved surfaces and recessed elements, that automated methods may not reliably capture. This benchmark, shown in Figure 5, served as the reference for identifying elements and checking the correspondence of extracted semantic classes. Architectural plans were additionally consulted to verify element counts and façade dimensions.

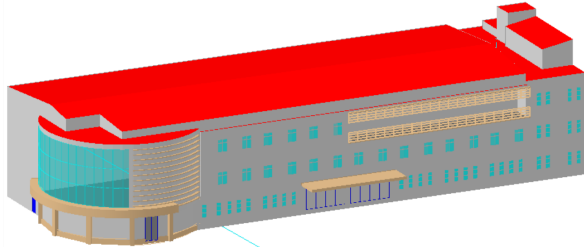


Figure 5. Manually constructed benchmark model

4.2 Geometric accuracy assessment

Geometric quality was evaluated using Cloud-to-Cloud (C2C) distance and Multiscale Model-to-Model Cloud Comparison (M3C2). The C2C metric provides a direct measure of point-wise proximity between the image-derived façade point cloud and the LiDAR reference. At the same time, M3C2 evaluates local surface agreement along estimated normals, enabling a more

surface-aware comparison of the two point clouds (Kharroubi et al., 2022). C2C comparison with the SLAM-based LiDAR reference shows that the façade points preserve the overall building form but lack fine-scale precision. The per-façade results of the C2C and M3C2 analyses are presented in Table 1.

Façade	C2C Mean distance	C2C Standard Deviation	M3C2 Distance
Front (South)	0.40 m	0.33 m	0.10 m
Back (North)	0.39 m	0.29 m	0.13 m
East	0.53 m	0.17 m	0.07 m
West	0.54 m	0.37 m	0.009 m
Overall	0.46 m	0.31 m	0.08 m

Table 1. C2C and M3C2 distances between omnidirectional and LiDAR-derived point clouds

The overall mean C2C distance of 0.46 m indicates moderate geometric agreement at the building scale. For LOD3 façade structuring, where planar surface consistency is more critical than millimeter-level precision, this level of deviation is sufficient to support geometric reconstruction of main façade planes, overhangs, and wall offsets. The mean M3C2 distance of 0.08 m further indicates that planar surfaces are generally well-captured, implying that most C2C discrepancies arise from fine-scale architectural detail rather than from systematic geometric errors. However, due to occlusions and limitations in camera placement, some areas were not fully reconstructed. For example, the upper portion of the east façade is missing in the image-derived reconstruction compared to the LiDAR reference due to tree occlusions.

4.3 Semantic accuracy assessment

The semantic performance of the K-Means-based HSV clustering approach was evaluated by comparing it with the benchmark model. Since the data are represented as point clouds, façade elements were quantified by clustering the corresponding detected features. Detection performance was measured by the correspondence between predicted and actual classes, while classification performance was evaluated using precision, recall, and F1 Score.

$$Precision = \frac{TP}{TP + FP}; Recall = \frac{TP}{TP + FN} \quad (5)$$

$$F1\ Score = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (6)$$

where TP represents correctly identified elements, FP represents incorrectly classified elements, and FN represents undetected reference elements.

Predicted labels were compared with ground-truth classes derived from the benchmark model. The resulting confusion matrix is shown in Table 2, and the derived performance metrics are summarized in Table 3.

		Predicted		
		Window	Door	Installation
Actual	Window	108	0	11
	Door	0	1	3
	Installation	2	2	17

Table 2. Confusion matrix of façade element classification.

Class	TP	FN	Precision	Recall	F1-score
Window	108	32	0.98	0.77	0.86
Door	1	2	0.33	0.33	0.33
Installation	19	0	0.58	1.00	0.70

Table 3. Detection and classification performance metrics for façade elements

The resulting semantic segmentation is shown in Figure 6, where façade elements are differentiated based on HSV properties, with windows, building installations, and doors represented as distinct point clusters. A total of 32 windows and 2 doors were not detected, primarily due to occlusions and limited façade visibility.

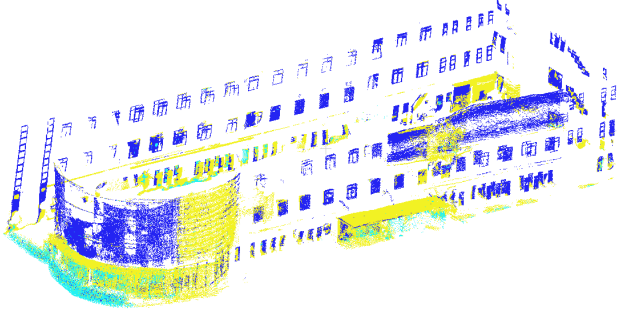


Figure 6. HSV-based semantic segmentation results, showing extracted façade elements including windows (blue), building installations (yellow), and doors (cyan).

Semantic performance is strongly influenced by visual distinctiveness and geometric separability. Window elements, which exhibit consistent color contrast and regular spatial patterns, achieved high classification performance despite moderate recall. Building installations showed strong detection performance but lower precision, indicating partial confusion with adjacent façade elements. Door elements were more difficult to recover due to limited visual contrast and their location near ground level, where occlusions, vegetation, and reduced point density affect reconstruction. Following minor manual refinement to reassign misclassified points, semantic feature detection completeness reached 81%, with the remainder comprising undetected elements. Overall, the results show that HSV-based segmentation can recover meaningful façade elements, although performance remains class-dependent and influenced by visibility, geometry, and color similarity.

4.4 Discussion

The results support the central premise of this study. A limited number of omnidirectional images can already provide meaningful façade geometry and semantic information, even prior to full LOD3 reconstruction. This is significant because omnidirectional imagery offers a wide field of view, enabling large portions of a building façade to be captured from relatively few acquisition points. Such efficiency is advantageous not only for field acquisition but also for future workflows that rely on sparse or publicly available street-level imagery. Further, we find that a limited number of omnidirectional images provides sufficient data for a realistic test case involving a building with complex geometry and partial occlusions.

At the same time, several limitations remain. Image-derived façade point clouds remain less geometrically precise than LiDAR references, particularly at fine scales. Semantic performance is class-dependent, with stronger results for visually distinct elements such as windows and weaker performance for classes with similar color appearances. These limitations are

consistent with early-stage workflows that rely on sparse, image-derived geometry.

The choice of HSV-based segmentation should be interpreted in the context of the input data. Unlike LiDAR-derived façade point clouds, which support geometry-driven segmentation through surface normal and planar constraints, façade point clouds reconstructed from sparse omnidirectional imagery exhibit uneven density and local geometric distortions introduced by monocular depth estimation. Under these conditions, a simple and interpretable color-based approach provides a practical baseline for semantic extraction. This approach remains a baseline and can be extended through more advanced classification strategies tailored to image-derived façade point clouds. Future work can build on this foundation by incorporating more advanced classifiers, contextual rules, or learning-based approaches tailored to image-derived façade point clouds.

Most importantly, the outputs presented in this study constitute the LOD3-ready geometric and semantic components required for full CityGML reconstruction. The LOD2 base model and the semantically segmented façade point cloud together provide the geometric and semantic foundation for CityGML-compliant LOD3 reconstruction. Semantic readiness, rather than complete semantic reconstruction, is proposed as an appropriate and achievable target for early-stage workflows that operate under constrained data acquisition, with the full reconstruction pipeline currently under active development. These results suggest that semantic readiness may represent a practical intermediate target between LOD2 reconstruction and full LOD3 completion in image-based urban modeling workflows operating under constrained acquisition conditions.

This positions Omni2LOD3 as a viable early-stage approach for LOD3 reconstruction in data-scarce environments, where traditional dense acquisition workflows are not feasible. Rather than treating sparse acquisition as a limitation, Omni2LOD3 reframes it as a design constraint under which LOD3-ready reconstruction remains achievable.

5. Conclusions and Ongoing Work

We demonstrate that LOD3-ready façade geometry and semantic information can be derived from sparse omnidirectional imagery without LiDAR or dense multiview image capture. Omni2LOD3 establishes a replicable minimal-data workflow that generates LOD3-ready components using accessible image-based inputs, specifically an LOD2 base model and a semantically segmented, geometrically representative façade point cloud that can support subsequent LOD3 refinement. In doing so, the study challenges the conventional assumption that dense acquisition is required for LOD3 reconstruction.

Geometric evaluation against SLAM-based terrestrial LiDAR confirms that the reconstructed façade point clouds capture the overall building form, including overhangs and balconies, even when generated from a limited number of omnidirectional images through monocular depth estimation. Semantic evaluation shows that HSV-based segmentation can recover façade elements with class-dependent performance, reaching 81% detection completeness after minor manual refinement. Semantic performance was class-dependent: windows were detected reliably due to their consistent color contrast, whereas doors proved difficult to recover because of their visual similarity to surrounding surfaces and ground-level occlusions.

Rather than achieving complete semantic reconstruction, the present results support semantic readiness as a practical target for early-stage, low-cost LOD3 workflows. In this context, the HSV-based method should be understood as a lightweight baseline for semantic extraction from sparse image-derived façade point clouds. More advanced image-based or learning-based classification methods may later be integrated into the broader Omni2LOD3 workflow to improve performance on visually ambiguous or architecturally complex façade elements. These findings open the possibility for scalable urban modeling workflows in settings where dense acquisition or LiDAR infrastructure remains inaccessible.

5.1 Ongoing Work

Current work focuses on completing the reconstruction pipeline by combining the LOD2 model with the semantically segmented façade point cloud to produce a CityGML 3.0-compliant LOD3 model. As shown in Figure 7, this involves deriving semantic surfaces and planar features from the generated point clouds, clustering semantically segmented points into façade elements, generating geometric proxies for openings and installations, and integrating these with the LOD2 envelope.

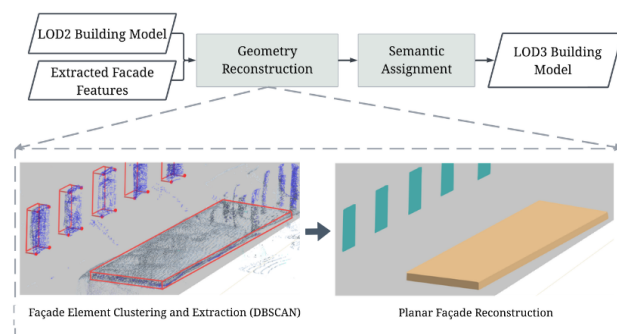


Figure 7. Conceptual overview of the ongoing Omni2LOD3 reconstruction pipeline

A particular proposed innovation in Omni2LOD3 is the handling of curved façade regions through tessellation or segmented approximation prior to feature reconstruction. In this way, the present study serves as a foundation for a more complete end-to-end image-based workflow that progresses from accessible data acquisition to more detailed, semantically meaningful 3D building models.

Building on this, ongoing work also focuses on improving semantic extraction through more advanced image-based and learning-based segmentation methods. In this context, the HSV-based approach serves as an initial step for semantic extraction, providing a simple and interpretable basis for façade segmentation. More advanced methods are expected to improve performance for visually ambiguous and architecturally complex façade elements.

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